Huawei Access Node (HEB-ACC-33) QoS Congestion Event (Microbursts, Interface Tail Drops, QoS Queue Starvation) Resolution SOP

Node: HEB-ACC-33 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_2887

Description

The failure domain of QoS congestion events on the Huawei access node (HEB-ACC-33) at Hebbal Edge is rooted in the intricacies of Quality of Service (QoS) policy enforcement and the interaction with the underlying 802.1Q-based Virtual LAN (VLAN) architecture. Specifically, this scenario is precipitated by the occurrence of microbursts, which are short-lived, high-intensity bursts of traffic that overwhelm the interface's buffering capacity, resulting in tail drops. Additionally, QoS queue starvation arises when high-priority traffic dominates the available bandwidth, leaving lower-priority traffic without sufficient resources. To exacerbate matters, the Huawei access node's implementation of 802.1Q prioritization and policing mechanisms can sometimes lead to unforeseen interactions between traffic flows, further contributing to congestion events. The combination of these factors necessitates a comprehensive approach to diagnosing and resolving QoS-related issues on this equipment.

Pre-requisite Checks

- Verify that the Hebbal Edge site is accessible and that the Huawei access node (HEB-ACC-33) is powered and online.

- Confirm that all redundant links are operational and that no network outages or interface issues are present.

- Review the QoS policy configuration on the Huawei access node to ensure accuracy and completeness.

Potential Root Causes

Cause	Probability
Misconfigured QoS policy leading to QoS queue starvation	High
Microbursts and interface tail drops due to insufficient interface buffering	Medium
Incorrectly set QoS policing parameters	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos policy, Expected Output: QoS policy configuration, including policing and shaping, Description: Verifies the accuracy and completeness of the QoS policy configuration on the Huawei access node.
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```
Command: display interface, Expected Output: Interface configuration and statistics, including buffering status, Description: Informs on the interface's buffering capacity and whether it is experiencing tail drops.

Command: display qos statistics, Expected Output: QoS statistics, including queue depths and drop rates, Description: Provides insight into the current QoS queue depths and drop rates, helping to identify potential congestion points.
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Resolution Steps

Step 1: Step: Verify and correct QoS policy configuration, Command: undo qos policy [policy-name], Description: Carefully review and correct the QoS policy configuration to prevent QoS queue starvation and ensure accurate policing and shaping.

Step 2: Step: Adjust interface buffering and policing parameters, Command: set interface [interface-name] buffer-size [buffer-size], Description: Increase the interface's buffering capacity to mitigate microbursts and interface tail drops, and adjust policing parameters to prevent QoS queue starvation.

Step 3: Step: Re-apply QoS policy with corrected policing parameters, Command: qos policy [policy-name] policing [policing-parameters], Description: Apply the corrected QoS policy with updated policing parameters to ensure accurate traffic management.

Post-Check Verification

```
Command: display qos policy, Expected Output: Corrected QoS policy configuration, Description: Verifies that the corrected QoS policy configuration is accurate and complete.

Command: display interface, Expected Output: Improved interface buffering status and reduced tail drops, Description: Informs on the improved interface buffering capacity and reduced tail drops.

Command: display qos statistics, Expected Output: Improved QoS queue depths and reduced drop rates, Description: Provides insight into the improved QoS queue depths and reduced drop rates, indicating successful resolution of congestion events.
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Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: qos policy [original-policy-name], Description: Restore the original QoS policy configuration to prevent unintended consequences of corrective actions.

- Step: Revert interface buffering and policing parameters, Command: undo set interface [interface-name] buffer-size [buffer-size], Description: Revert the interface's buffering capacity and policing parameters to their original settings.